NORFOLK NAVAL STATION
Norfolk, VA

Section 3.1 - Q47 Fishing Pier

Contract: N62583-12-D-0749 Inspection Date: 07/15/2013
Contractor: Collins Engineers, Inc.
123 N. Wacker Drive, Suite 900
Chicago, Illinois 60606
Facility: Q47 Fishing Pier **iNFADS:** NFA100001159293 **PRN:** 200022
Location: 36.963138, -76.321799 **Estimated Repair Cost*:** $ 11,187
**Repair cost includes Design Allowances, Contractor Overhead & Profit, and Inflation Allowances.*
See Appendix C - Cost Estimate for detailed analysis.

Condition Index (CI)	**75**	**Max Water Current**	0.5kn
Engineering Assessment Rating	Satisfactory	**Water Clarity**	3ft
Operational Rating	C1	**Tide Variation**	4ft
5 Year Projected CI	74	**Max Water Depth**	14ft
10 Year Projected CI	73	**Seasonal Water Temp**	75°F
Year(s) Previously Inspected	2007	**Seasonal Ambient Temp**	88°F

Facility Usage Description:
The Q47 Fishing Pier is part of the Morale, Welfare and Recreation (MWR) program of Naval Station Norfolk. The pier is located approximately 1,000 ft east of Pier 14. The pier is approximately 283 ft long and extends to the north from the shore. The approach is approximately 240 ft long by 10 ft wide, and the main pier head is approximately 43 ft long by 40 ft wide with a protective canopy covering the majority of the pier head. A waterline extends to the pier head from the shore, and lighting is provided by six high mast light poles and one light fixture at the protective canopy. The pier is accessible to pedestrians only.

Summary of Repair Recommendations:
It is recommended that the concrete cracking along the bottom of the double tee beams be repaired. It is also recommended that the steel roof of the protective canopy be cleaned and recoated and that the failed support brackets for the waterline be repaired.

Impact To Mission If Repairs Not Provided:
If repairs are not made to the double tee beams, the deterioration in the webs will continue to progress at an accelerated rate now that the cracking has exposed the prestressing strands to the corrosive marine environment. Likewise, the corrosion on the steel support members for the roof will continue to progress as the protective coatings have begun to fail. For the waterline, if the deficient support brackets are not replaced, the line will fail soon and the fresh water supply to the pier will be interrupted.

Operational Restrictions:
There are currently no operational restrictions to the Q47 Fishing Pier.

Additional Facility Photos

Photo 3.1-1: Overall view of the Q47 Fishing Pier, looking northwest.

Photo 3.1-2: Overall view of the Q47 Fishing Pier, looking northeast.

H1010 - Substructure **CI: 85**

H101001 - Pile Foundations **CI: 85**

UFII Component	H10100101	Piles Asset Type(s): PB
Findings	The piles are in Good Condition. There are 18 battered piles and no vertical piles supporting the pier. Pile 4F exhibited a small area of section loss that is 8 in. high by 3 in. wide with up to 1 in. of penetration and no exposed reinforcing steel. There are no other notable defects on the concrete piles.	
Recommendations	No repairs are recommended at this time. The area of section loss noted at Pile 4F should be monitored during future inspections for any further deterioration.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H10100101_PB

Photo 3.1-3: View of the concrete battered piles at Bent 1, looking southwest.

H1010 - Substructure **CI: 85**

H101002 - Pile Caps **CI: 90**

UFII Component	H101002	Pile Caps Asset Type(s): PC & PC1
Findings	The pile caps are in Good Condition. There are five concrete piles caps, three are 9.7 ft long and two are 39.7 ft long. There are no notable defects on the concrete pile caps.	
Recommendations	No repairs are recommended at this time.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H101002_PC

Photo 3.1-4: View of the concrete pile cap at Bent 2, looking north.

H1020 - Superstructure **Repair Cost $8,839** **CI: 75**

H102001 - Beams and Girders **Repair Cost $6,106** **CI: 75**

UFII Component	H102001	Beams and Girders Asset Type(s): B, B1, & B2
Findings	The beams are in Satisfactory Condition overall. There are four prestressed double tee concrete beams spanning between Bents 0 and 4 and four prestressed double tee beams spanning between Bents 4 and 5. There are four 18 in. diameter reinforced concrete beams supporting the steel roof. Six of the double tee beams exhibit horizontal cracking along the bottom face of the webs. The cracking ranges from 1/8 in. wide to 1/2 in. wide. There are no notable defects for the beams supporting the steel roof.	
Recommendations	It is recommended that the areas of concrete cracking and associated deterioration noted along the double tee beam webs be repaired.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H102001_B

Photo 3.1-5: View of the concrete double tee beam between Bents 1 and 2, looking north.

Photo Type: Defect UFII/Asset Type: H102001_B

Photo 3.1-6: View of the 1/2 in. wide crack in the double tee beam north of Pile 1C, looking southwest.

Photo Type: Defect UFII/Asset Type: H102001_B

Photo 3.1-7: View of the 1/4 in. wide crack in the bottom of the double tee beam north of Pile 1D, looking south.

Photo Type: Typical UFII/Asset Type: H102001_B2

Photo 3.1-8: View of the concrete beams supporting the steel roof, looking northwest.

H1020 - Superstructure **Repair Cost $8,839** **CI: 75**

H102002 - Columns **CI: 90**

UFII Component	H102002	Columns Asset Type(s): COL
Findings	The columns are in Good Condition. There are four 18-in.-diameter concrete columns supporting the steel roof. There are no notable defects on the concrete columns.	
Recommendations	No repairs are recommended at this time.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H102002_COL

Photo 3.1-9: View of the concrete column condition, looking northwest.

H1020 - Superstructure **Repair Cost $8,839** **CI: 75**

H102004 - Other Superstructure Components **Repair Cost $2,733** **CI: 75**

UFII Component	H102004	Other Superstructure Components Asset Type(s): ROOF
Findings	The steel roof is in Satisfactory Condition. The steel support members exhibit areas of coating failure with minor corrosion on 5 percent of the surface areas.	
Recommendations	It is recommended that the steel members be cleaned and recoated.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H102004_ROOF

Photo 3.1-10: View of an area of coating failure and minor surface corrosion on the steel roof, looking southwest.

H1050 - Appurtenances **CI: 85**

H105001 - Handrails **CI: 85**

UFII Component	H105001	Handrails Asset Type(s): HRL
Findings	The handrails are in Good Condition. There is 645 linear ft of handrail along the perimeter of the pier. The steel support members are properly secured to the tee beams, and the timber rails exhibit only minor to moderate weathering.	
Recommendations	No repairs are recommended at this time.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H105001_HRL

Photo 3.1-11: View of the handrail condition, looking northwest.

H1050 - Appurtenances **CI: 85**

H105006 - Life Rings **CI: 90**

UFII Component	H105006	Life Rings Asset Type(s): LRH
Findings	The life rings and life ring supports are in Good Condition. There are three life rings and three associated life ring supports, all of which appear to be in proper operational condition.	
Recommendations	No repairs are recommended at this time.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H105006_LRH

Photo 3.1-12: View of the typical life ring condition, looking northeast.

H5010 - Civil/Mechanical Utilities **CI: 75**

H501001 - Potable Water **CI: 75**

UFII Component	H50100101	Pipes & Fittings Asset Type(s): LPW
Findings	The potable waterline is in Satisfactory Condition. There is 220 linear ft of insulated potable waterline along the western side of the pier. The waterline appears to be in operational condition. The supports between Bents 0 and 1, however, are failed with the waterline sagging up to 2 ft and "makeshift" ropes securing the waterline at only two locations.	
Recommendations	It is recommended that the five failed supports between Bents 0 and 1 be replaced so that the waterline is properly supported.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H50100101_LPW

Photo 3.1-13: View of the insulated waterline condition, looking southeast.

Photo Type: Defect UFII/Asset Type: H50100101_LPW

Photo 3.1-14: View of the "makeshift" rope supports and sagging waterline between Bents 0 and 1, looking northeast.

H5020 - Electrical Utilities **Repair Cost $2,349** **CI: 90**

H502001 - Electrical Power Distribution **Repair Cost $2,349** **CI: 90**

UFII Component	H50200103	Cables Asset Type(s): LPWR
Findings	The electrical conduits are in Good Condition. There is 334 linear ft of PVC conduit extending along the western side of the pier along the deck topside, and there is 250 linear ft of steel conduit extending along the western side of the pier along the deck underside. The entire conduit is securely attached with no significant deterioration or damage noted.	
Recommendations	No repairs are recommended at this time.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H50200103_LPWR

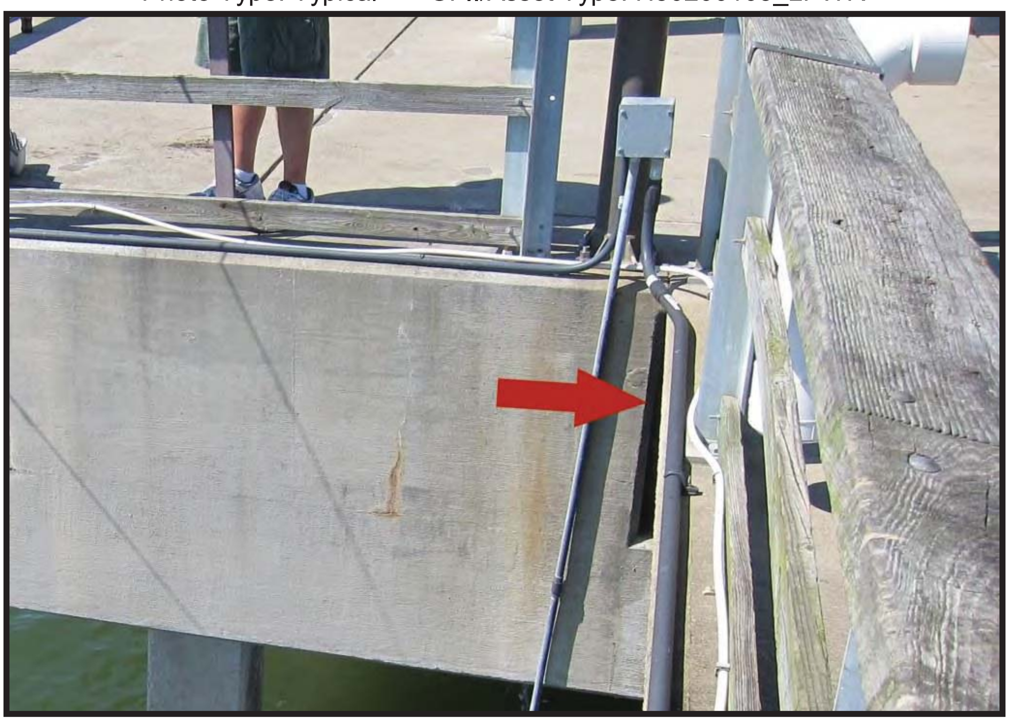


Photo 3.1-15: View of the PVC conduit, looking north.

Photo Type: Typical UFII/Asset Type: H50200103_LPWR

Photo 3.1-16: View of the steel conduit, looking northeast.

H5020 - Electrical Utilities **Repair Cost $2,349** **CI: 90**

H502003 - Lighting **CI: 90**

UFII Component	H502003	Lighting Asset Type(s): LGT, LGT1, & LGT2
Findings	The lights are in Good Condition. The six lights and associated support poles appear to be in proper operational condition with no significant defects or deterioration noted. The navigational light at the outboard end of the pier and the light attached to the steel canopy appear to be in proper operational condition with no significant defects or deterioration noted.	
Recommendations	No repairs are recommended at this time.	

Supporting Photos

Photo Type: Typical UFII/Asset Type: H502003_LGT

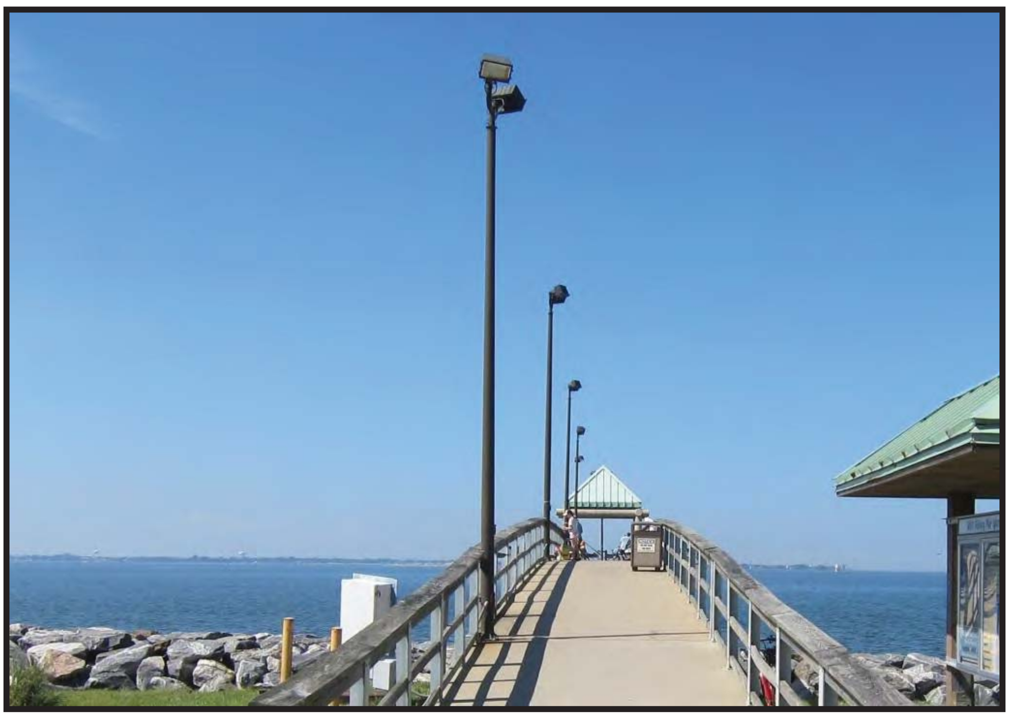


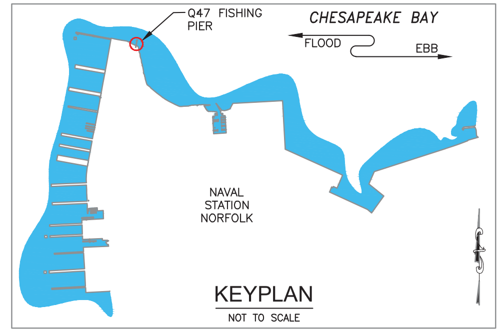
Photo 3.1-17: View of the lighting fixtures and support poles, looking north from the shore.

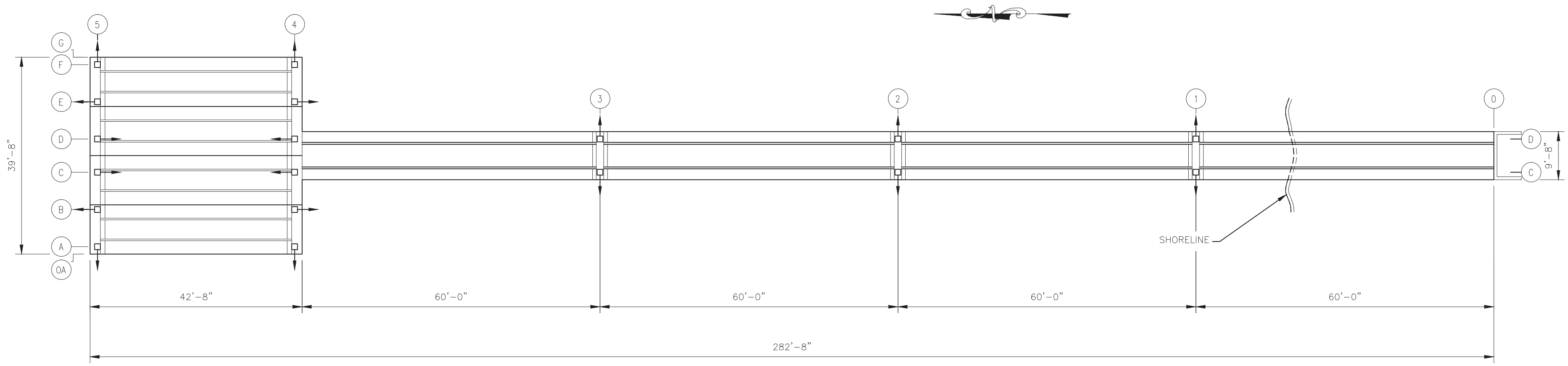
Photo Type: Typical UFII/Asset Type: H502003_LGT1

Photo 3.1-18: View of the navigational light at the outboard end of the pier, looking south.

Photo Type: Typical UFII/Asset Type: H502003_LGT2

Photo 3.1-19: View of the light attached to the steel canopy.





PLAN
SCALE: 1"=20'

COLLINS ENGINEERS INC
COLLINS ENGINEERS, INC.
123 NORTH WACKER DRIVE, SUITE 900
CHICAGO, IL., 60606

GRAPHIC SCALE
0 20 40 FT.
1"=20'

NAVFAC Naval Facilities Engineering Command

DATE: MARCH 2014

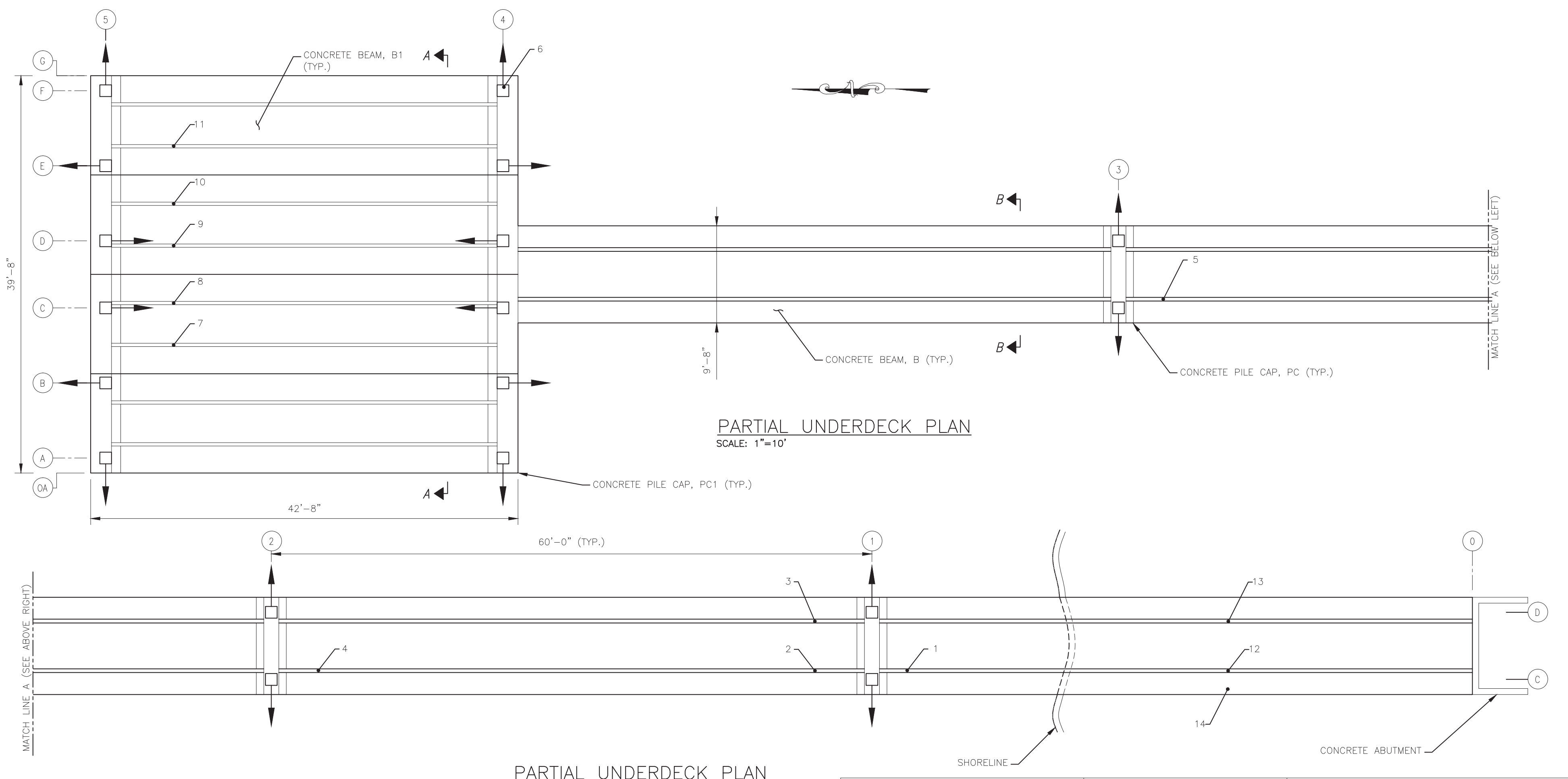
CONTRACT NUMBER
N62583-12-D-0749
Delivery Order No. 0009

NAVAL FACILITIES ENGINEERING COMMAND
ENGINEERING AND EXPEDITIONARY WARFARE CENTER
WASHINGTON, D.C.

NAVAL STATION NORFOLK | NORFOLK, VA | FIG. NO.

Q47 FISHING PIER
FACILITY PLAN

3.1-1



LEGEND:

12 DEFECT LOCATION AND IDENTIFICATION NUMBER (SEE APPENDIX B-STRUCTURAL DATA TABLE FOR MORE INFORMATION)

□ INDICATES SQUARE CONCRETE PILE, P

□→ INDICATES SQUARE CONCRETE BATTERED PILE, PB

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GRAPHIC SCALE
0 10 20 FT.
1"=10'

NAVFAC Naval Facilities Engineering Command

DATE: MARCH 2014

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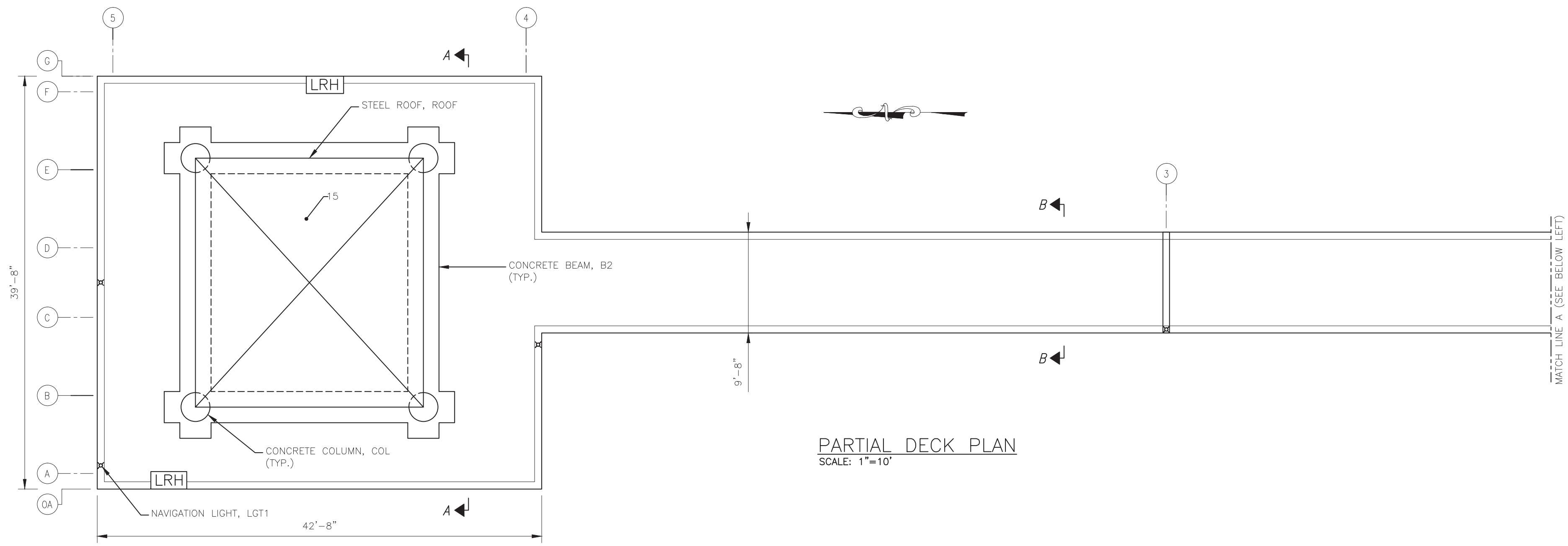
WASHINGTON, D.C.

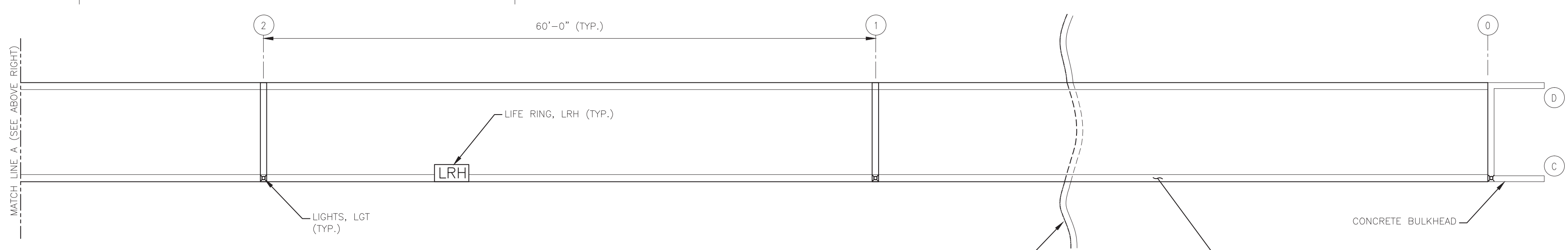
NAVAL STATION NORFOLK — NORFOLK, VA

Q47 FISHING PIER
UNDERDECK PLAN

FIG. NO. **3.1-2**



PARTIAL DECK PLAN
SCALE: 1"=10'



PARTIAL DECK PLAN
SCALE: 1"=10'

LEGEND:

12 — DEFECT LOCATION AND IDENTIFICATION NUMBER (SEE APPENDIX B–STRUCTURAL DATA TABLE FOR MORE INFORMATION)

LRH — INDICATES LIFE RING, LRH

ASSET(S) NOT SHOWN

LGT2– LIGHT

LPWR–POWER DISTRIBUTION LINES

LPW–POTABLE WATER LINE

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CHICAGO, IL., 60606

GRAPHIC SCALE
0 10 20 FT.
1"=10'

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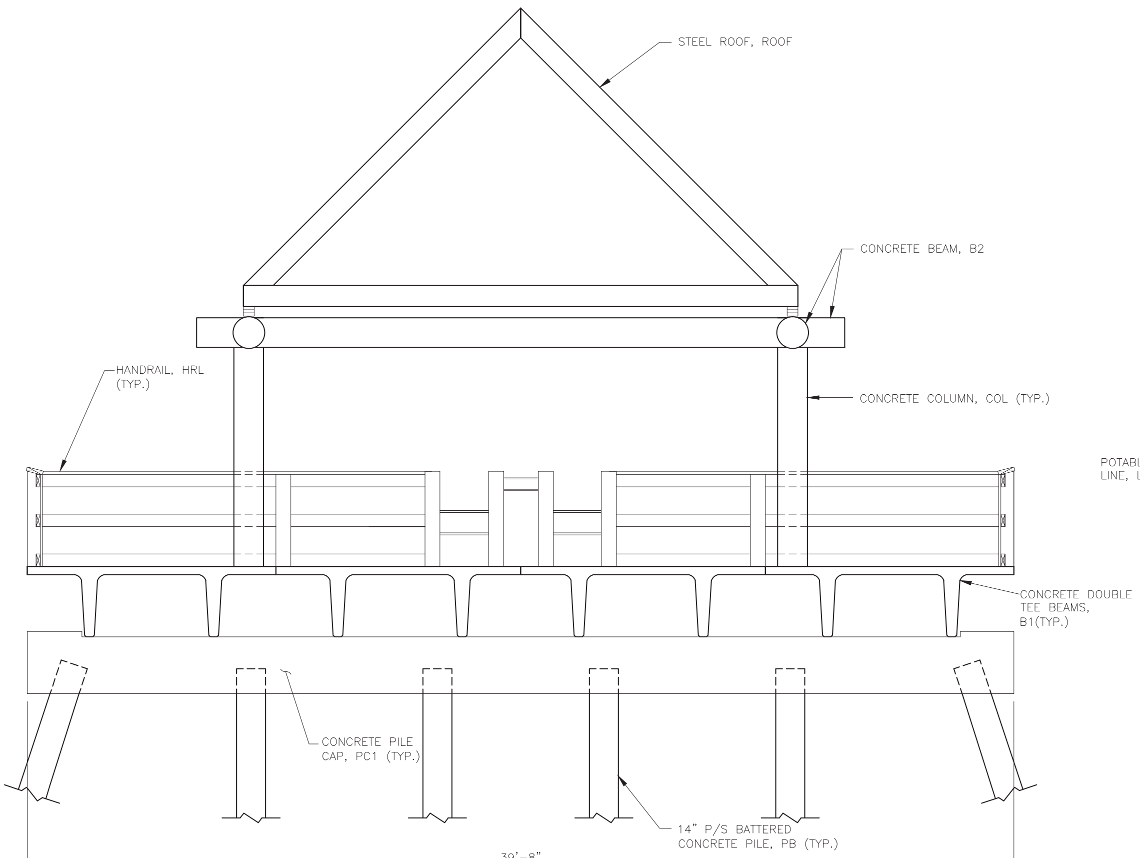
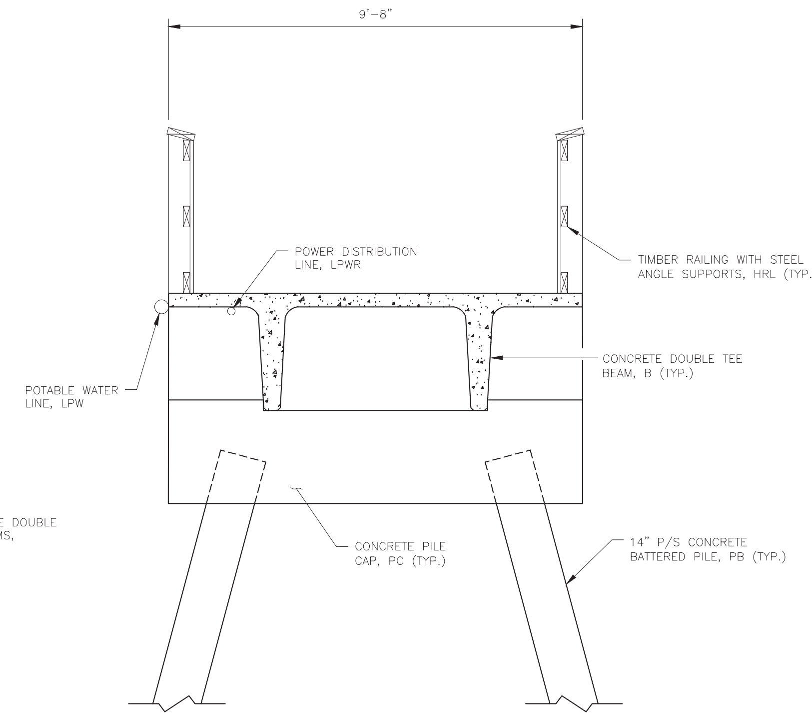
Q47 FISHING PIER
DECK PLAN

FIG. NO. 3.1-3



SECTION A–A
SCALE: 1"=5'



SECTION B–B
SCALE: 1"=3'

COLLINS ENGINEERS, INC.
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123 NORTH WACKER DRIVE, SUITE 900
CHICAGO, IL, 60606

GRAPHIC SCALE
0 3 6 FT.
1"=3'
0 5 10 FT.
1"=5'

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Q47 FISHING PIER
TYPICAL SECTIONS

FIG. NO. 3.1-4